

Project Report

“CFD – Simulation for Energy Systems”

CFD for Methane Combustion: the effect of FGR and Bayesian Optimization on NO production

submitted by: Group 10

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Summary

This report presents a Computational Fluid Dynamics (CFD) simulation of a steady-state, non-premixed methane-air diffusion flame within a 2D axisymmetric cylindrical combustor. Utilizing the $k - \omega$ SST turbulence model alongside the Eddy-Dissipation Model, this study captures the interactions among turbulent momentum transport, convective heat transfer, and rapid chemical reactions. A mesh-refinement study has shown negligible sensitivity to grid resolution, establishing a reliable computational domain for subsequent emission modeling. The baseline simulation accurately resolves the stoichiometric mixing boundary, highlighting a localized peak combustion temperature of approximately 2300 K, with a mass-weighted average outlet temperature of 1840 K. Moreover, the evaluation of nitrogen oxide (NO) formation, conducted through both thermal and prompt mechanisms, indicates a baseline exit emission level of 5067ppm, primarily influenced by the high-temperature thermal pathway. The application of Flue Gas Recirculation (FGR) is explored as a proactive emission control strategy, demonstrating that incorporating a 473 K recycle stream effectively dilutes the reacting mixture and mitigates peak temperatures. This approach achieves nearly a 99% reduction in total NO emissions, decreasing them to 52 ppm. Finally, the thermal NO emissions were further reduced by 99% by optimizing input parameters $x = [v_{air}, v_{FGR}, T_{FGR}]$ using Bayesian Optimization. The resulting thermal NO outlet concentration was 0.033 ppm. In summary, this study not only affirms the effectiveness of FGR in significantly reducing pollutant formation in combustion, but also shows how BO with 23 data points can be used to further optimize the system for minimum NO_x production.

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Formula Symbols and Abbreviations

Latin characters

Symbol	Description	Unit
A	Eddy-dissipation model reactant-rate constant	–
B	Eddy-dissipation model product-rate constant	–
c	Speed of sound	m s^{-1}
d	Characteristic diameter	m
d_h	Hydraulic diameter	m
$f(\mathbf{x})$	Unknown system response in Gaussian Process regression	–
$f_{\min}$	Best feasible objective value used in Expected Improvement	–
k	Turbulent kinetic energy	$\text{m}^2 \text{s}^{-2}$
$k(\mathbf{x}, \mathbf{x}')$	Gaussian Process covariance kernel	–
Ma	Mach number	–
N	Number of observations or training samples	–
Re	Reynolds number	–
T	Temperature	K
T_{FGR}	Flue-gas-recirculation inlet temperature	K
T_{out}	Mass-weighted-average outlet temperature	K
T_{baseline}	Baseline mass-weighted-average outlet temperature	K
$T_{\min}$	Minimum allowable outlet temperature	K
v	Velocity component	m s^{-1}
v_{air}	Air inlet velocity	m s^{-1}
v_{FGR}	Flue-gas-recirculation inlet velocity	m s^{-1}
X_i	Mole fraction of species i	–
Y_i	Mass fraction of species i	–
$\mathbf{x}$	Bayesian Optimization input/design vector	–
y_i	Observed training output	–

Greek characters

Symbol	Description	Unit
$\alpha(\mathbf{x})$	Constrained acquisition function	–
ε	Turbulent dissipation rate	$\text{m}^2 \text{s}^{-3}$
γ	Gaussian Process kernel amplitude hyperparameter	–
λ	Thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$
μ	Dynamic viscosity	Pa s
μ^*	Gaussian Process posterior mean	–
μ_{NO_x}	Predicted mean outlet NO_x mass fraction	–
ν	Kinematic viscosity	$\text{m}^2 \text{s}^{-1}$
ν_t	Turbulent kinematic viscosity	$\text{m}^2 \text{s}^{-1}$
ρ	Density	kg m^{-3}
σ	Standard deviation	–
σ_n	Gaussian Process observation-noise standard deviation	–
Σ^*	Gaussian Process posterior covariance	–
$\Phi(\cdot)$	Standard normal cumulative distribution function	–
ω	Specific turbulence dissipation rate	s^{-1}

Abbreviation	Description
BO	Bayesian Optimization
CDF	Cumulative Distribution Function
CFD	Computational Fluid Dynamics
CPU	Central Processing Unit
EDM	Eddy-Dissipation Model
EI	Expected Improvement
FGR	Flue Gas Recirculation
GP	Gaussian Process
GPR	Gaussian Process Regression
iid	Independent and identically distributed
MWA	Mass-Weighted Average
NO_x	Nitrogen oxides
ppm	Parts per million
RANS	Reynolds-Averaged Navier–Stokes
SST	Shear Stress Transport

1 Introduction

1.1 Background of Combustion

Combustion is a fundamental physicochemical process defined by a rapid exothermic reaction between a fuel and an oxidizer, usually oxygen from the air [1]. This highly energetic reaction releases significant heat and forms the backbone of the global energy infrastructure [2, 3]. Despite the swift progress in renewable technologies, combustion continues to be the primary method for power generation, aerospace propulsion, and heavy industrial manufacturing.

The physics of combustion involves a complex interaction of turbulent fluid dynamics, multiple modes of heat transfer, and chemical kinetics [1, 2]. The key principles rely on the efficient transport of chemical species: reactants must be well mixed, heated to ignition temperatures, and then released as combustion products.

1.2 Computational Fluid Dynamics and Ansys Fluent

Historically, designing and optimizing combustion chambers required costly and time-consuming physical prototypes. However, the rise of Computational Fluid Dynamics (CFD) has transformed this process, allowing for the simulation of complex reacting flows in a virtual environment [4, 5]. By solving the governing equations for mass, momentum, energy, and chemical species, CFD enables engineers to visualize flow fields and thermodynamic behaviors that are difficult to measure.

Ansys Fluent is a leading CFD software tool, widely used for modeling fluid and thermal phenomena [6]. It offers a robust platform for handling the complexities of turbulence-chemistry interactions, making it an essential resource for engineers seeking innovation and precision in their designs.

2 Description of Project Task and Motivation

This report presents a steady-state CFD simulation for methane combustion in a turbulent diffusion flame furnace. CFD allows for investigation of the coupled turbulent flow, heat release and mass transport without solely relying on expensive experimental measurements.

The goal of this project is to model and evaluate the methane combustion process in ANSYS Fluent through examining contour plots for temperature and species mass fractions along the reactor. Meshing is evaluated by comparing the outlet temperature with different mesh resolutions. NO formation is simulated and compared across thermal, prompt and combined mechanisms. Finally, Flue Gas Recirculation stream is added to the system and its influence on NO production is assessed. The NO production is further reduced by optimizing the FGR inlet conditions.

Figures 1 and 2 show the exact dimensions of the 2D-axisymmetric combustor geometries. The outlet temperature and pressure refer to the backflow values. The actual outlet temperature is computed with ANSYS Fluent.

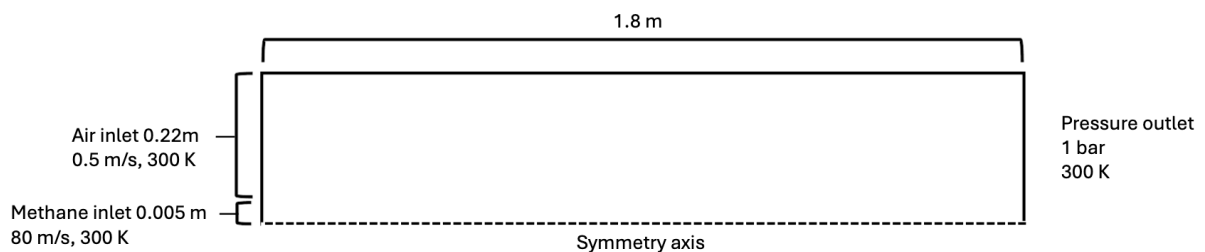


Figure 1: Geometry and dimensions of the combustor. The values at the pressure outlet correspond to the backflow boundary conditions.

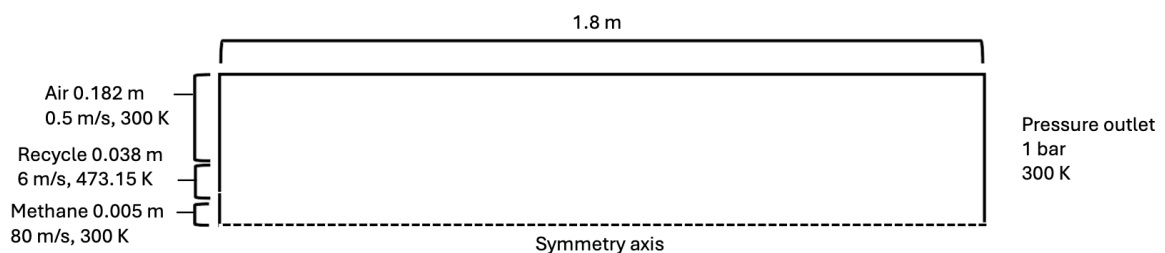


Figure 2: Geometry and dimensions of the combustor with FGR.

3 Theoretical background

3.1 Dimensionless Numbers and Flow Regime Justification

The Reynolds number is defined as the ratio of inertial forces to viscous forces:

$$Re = \frac{\rho v d}{\mu} \quad (3.1)$$

where ρ represents the fluid density in kg/m^3 , v denotes the inlet velocity in m/s , d indicates the characteristic length in m (such as nozzle diameter or hydraulic diameter), and μ stands for the dynamic viscosity in Ns/m^2 . To determine the flow regime (either laminar or turbulent) and set up the Ansys Fluent solver correctly, the Reynolds number (Re) must be calculated for both the central methane jet and the annular air co-flow.

1. Methane Fuel Jet: Given the standard properties of methane at a temperature of 300 K, the density is approximately $\rho \approx 0.656 \text{ kg/m}^3$ and the dynamic viscosity is $\mu \approx 1.1 \times 10^{-5} \text{ kg/(m} \cdot \text{s)}$ [7].

For the project specifications, with an inlet velocity of $v = 80 \text{ m/s}$ and a nozzle diameter of $d = 0.01 \text{ m}$, the Reynolds number Re_{CH_4} can be calculated as follows:

$$Re_{CH_4} = \frac{0.656 \cdot 80 \cdot 0.01}{1.1 \times 10^{-5}} \approx 47,700$$

2. Air Co-flow: Assuming standard properties for air at a temperature of 300 K, the density is approximately $\rho \approx 1.177 \text{ kg/m}^3$ and the dynamic viscosity is $\mu \approx 1.85 \times 10^{-5} \text{ kg/(m} \cdot \text{s)}$ [8]. The hydraulic diameter for the outer annulus is calculated as follows:

$$d_h = d_{outer} - d_{inner} = 0.45 \text{ m} - 0.01 \text{ m} = 0.44 \text{ m}.$$

With an inlet velocity of $v = 0.5 \text{ m/s}$, the Reynolds number is calculated as:

$$Re_{air} = \frac{1.177 \cdot 0.5 \cdot 0.44}{1.85 \times 10^{-5}} \approx 14,000.$$

Flow Regime Justification: The Reynolds numbers for the methane jet (around 47,700) and

the air co-flow (approximately 14,000) are high, indicating that a turbulent flow regime governs the fluid dynamics within the combustor [6]. To model the turbulent mixing, momentum transport, and complex shear layers characteristic of this diffusion flame, a RANS turbulence closure equations, specifically the $k - \omega$ Shear Stress Transport (SST) model, is used.

3.2 Mach Number and Compressibility

The Mach number (Ma) assesses compressibility effects by comparing the local flow velocity (v) to the local speed of sound (c) in the fluid medium [9].

$$Ma = \frac{v}{c} \quad (3.2)$$

1. Methane Fuel Jet: In the case of the central methane jet, which enters the domain at a velocity of 80 m/s and an initial temperature of 300 K, the estimated local speed of sound is around 449 m/s [9], resulting in a maximum Mach number of:

$$Ma_{CH_4} = \frac{80}{449} \approx 0.178$$

2. Air Co-flow: In the case of the slow-moving co-flow air entering at 0.5 m/s and 300 K, the speed of sound is approximately 347 m/s [9], leading to Mach number of

$$Ma_{air} = \frac{0.5}{347} \approx 0.0014$$

In the field of fluid dynamics, flows characterized by a Mach number below 0.3 are classified as incompressible, as density fluctuations due to dynamic pressure changes are negligible [5]. Therefore, density is still temperature and composition dependent. Given that the maximum Mach number at the inlet is 0.178, which is well below this threshold, the flow can be treated as incompressible. As a result, a pressure-based solver in ANSYS Fluent is used for simulations.

3.3 Chemical Reactions and Species Transport

The primary heat release during the combustion of methane is governed by a singular global reaction representing its complete oxidation:



By operating at a fuel-lean equivalence ratio of 0.76 and incorporating 28% excess air, atmospheric nitrogen (N_2) plays a crucial role as a thermal diluent. It effectively absorbs heat, thus helping to mitigate peak flame temperatures [1]. To model the complex interactions between turbulence and chemical reactions, we employ the Eddy-Dissipation Model (EDM). The Eddy Dissipation Model (EDM) effectively calculates the primary heat-release reactions, while trace nitrogen oxide (NO_x) emissions are determined through a decoupled post-processing step. This approach allows for evaluation of NO_x formation through different mechanisms by isolating two significant finite-rate chemical pathways: the Thermal (Zeldovich) mechanism (See Eqs. 3.4-3.6), prevalent in high-temperature hotspots, and the Prompt (Fenimore [10]) mechanism (See below) [11].



4 Models and Materials - Task a)

The 2D-axisymmetric combustor is modeled with a pressure-based steady-state model, with absolute velocity formulation. Energy equation is enabled to calculate conductive and convective heat transfer [12] and to approximate the axial temperature distribution along the reactor. For the scope of this project, radiative heat transfer is not included.

4.1 Turbulence Model - k - ω SST

Since Re for both methane and air inlet streams is very high ($\approx 47,700$ and $\approx 14,000$, respectively), the flow profile in the combustor is treated as turbulent. The turbulent flow is modeled using Reynolds Averaged Navier Stokes (RANS) equations closed with eddy viscosity model [6].

As the non-premixed methane enters the confined combustor at high speed through a small nozzle, both free shear like conditions and boundary layer effects near the wall must be considered. Therefore, k - ω Shear Stress Transport (k - ω SST) turbulence model is used where the k - ω model is dominating near the wall and the k - ϵ model is dominating near the free shear region [13]. The k - ω model accounts for turbulent kinetic energy k and specific turbulent dissipation ω (Eq. 4.1), that are solved with two additional differential equations. The models are solved using the default model constants (See Appendix 7).

$$k\text{-}\omega: \quad \nu_t = \frac{k}{\omega} \qquad k\text{-}\epsilon: \quad \nu_t = \frac{k^2}{\epsilon} \qquad (4.1)$$

4.2 Materials

Species transport model models the chemical reaction in the methane-air mixture. The reaction occurs throughout the combustor and is treated as volumetric. Physical properties for the species present in the methane-air mixture are retrieved from *methane-air* in the Mixture Material database of ANSYS Fluent. The pre-defined *methane-air* mixture contains CH_4 , O_2 , CO_2 , H_2O and N_2 .

Due to the fast chemical kinetics compared to the rate of mixing by turbulent flow, Eddy Dissipation Model (EDM) is used for the turbulence-chemistry interaction. The EDM ignores

the Arrhenius rate determined reaction kinetics and instead defines the reaction rate using the turbulence parameters $\frac{k}{\epsilon}$, reactant concentration and coefficients A ($=4.0$) and B ($=0.5$) to scale the reaction rates for reactants and products, respectively [6]. As the Mach number ($Ma = 0.178$) is less than 0.3, the density properties of the fluid mixture are set to *incompressible-ideal-gas*.

4.3 Boundary Conditions

The combustor is modeled as a 2D axisymmetric geometry with a length of 1.8 m and a diameter of 0.45 m , with an internal methane nozzle with a diameter of 0.01 m (See Fig. 1).

Physical boundary conditions are applied at the air and methane inlets as Dirichlet boundary conditions by specifying constant velocity, temperature and species composition. The axial air and methane inlet velocities are set to 0.5 m/s and 80 m/s , respectively, with both streams entering at 300 K . The oxygen mass fraction of the air stream is 0.23, while the methane stream is pure methane. The turbulent intensity is set to 10 % for both inlets, describing the relative velocity fluctuations [14].

The combustor wall temperature is set to 300 K. The ANSYS provided gascomb.msh geometry [6] contains an additional adiabatic wall next to the methane inlet nozzle, that is set to 0 W/m^2 heat flux. No-slip ($u_{x_{wall}} = 0$) and impermeability conditions ($u_{y_{wall}} = 0$) are applied on all walls. Neither the refined meshes nor the FGR geometry contains the wall adjacent to the methane nozzle. As a result, also the total cell count with gascomb.msh geometry is lower at 1615 cells, compared to 1624 cells with the recreated geometry and mesh.

The combustor outlet pressure is set to 1 bar, with a 0 Pa Gauge pressure. The backflow turbulent intensity and temperature are set to 10 % and 300 K and the O_2 mass fraction is set to 0.23. Backflow values are only utilized when backflow occurs at the outlet [15].

5 Results

The simulations were conducted using ANSYS Fluent and default solver settings. The residual convergence criteria were set to 10^{-3} for continuity, velocities, turbulence equations, and species equations, and 10^{-6} for the energy equation. The solution was initialized using a hybrid method with a time-scale factor of 5.

Simulation results are first analyzed for the gascomb geometry provided by ANSYS [6], using contour plots of temperature, species concentrations, and the velocity vector field. Mesh refinement and NO_x analyses were performed using a recreated geometry and mesh. The lateral cell counts were retained, but the meshing bias, total cell count, and methane inlet wall boundary conditions (See Section 4.3) were not reproduced exactly. Comparing to the recreated meshing, the total cell count with the provided gascomb mesh is lower by 9 cells. Additionally, there is no adiabatic wall adjacent to the methane inlet. The smaller total cell count does not necessarily indicate smaller control volumes, as the total area of the 2D-axisymmetric combustor is also different with gascomb due to the empty space in between the air and methane inlets (See Appendix 7). The meshing bias used in gascomb.msh could not be recovered, and a similar-looking meshing was produced with bias factor of 8.0 applied on the vertical walls and 35.0 on horizontal walls. Differences in meshing bias can have an impact on the simulation results, as the bias changes the local size of the control volumes in the region of high concentration gradients, where the bias factor is used to refine the mesh (inlet wall and symmetry axis).

5.1 Contour plots - Task b)

5.1.1 Temperature

According to the static temperature contour plot, the temperature hotspots approaching a peak value of 2300 K are mainly located along the turbulent shear layer, where the high-velocity central methane jet intersects with the surrounding low-velocity co-flow air. The temperature is overestimated as the effects of radiation are neglected. In this system, the importance of radiation is almost equal to importance of convection, as shown by the ratio of 1.09 for convective to radiative heat transport as calculated in ANSYS problem specification [6].

In this non-premixed diffusion flame, combustion occurs where fuel and oxygen mix in stoi-

chiometric proportions, creating an expanding cone that appears as a diagonal band in the 2D axisymmetric plane, just downstream of the methane inlet nozzle. The core of the methane jet remains cool at the inlet due to insufficient mixing with the oxidizer. Because the combustor operates with excess air, this unreacted air acts as a thermal diluent, effectively absorbing heat from the reaction zone [1]. Meanwhile, the reactor wall maintains a stable temperature of 300 K, governed entirely by the established isothermal boundary condition.

The surface integrals at the domain outlet determined with ANSYS Fluent indicate a mass-weighted average exit temperature of approximately 1840 K. As illustrated in Figure 3, the exhaust region maintains a significantly elevated thermal profile up to the exit boundary. This bulk exit temperature of 1840 K is lower than the absolute peak stoichiometric flame temperature primarily due to volumetric thermal dilution rather than boundary heat loss. According to the simulated energy flux reports, the total heat released by the combustion reaction is approximately 204.4 kW, while the heat transfer to the 300 K isothermal outer wall accounts for a loss of only 13.2 kW (roughly 6.5% of the total heat release). Because boundary heat extraction is minimal, the vast majority of the thermal energy remains within the fluid, where it is absorbed by the surrounding excess air as the flow mixes and convects downstream.

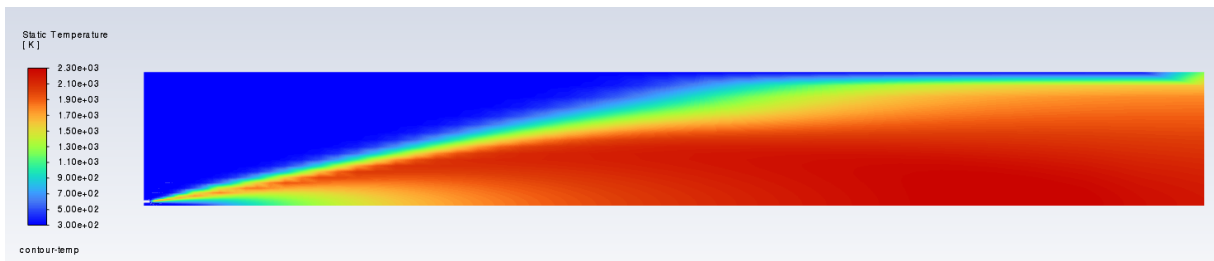


Figure 3: Contour plot of static temperature (K) across the 2D axisymmetric gas combustor domain, highlighting the primary reaction zone and peak temperature hotspots reaching approximately 2300 K.

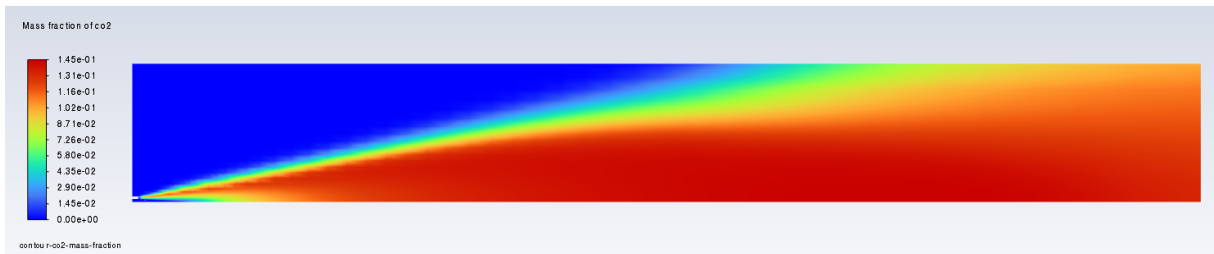
5.1.2 Species Mass Fractions

The species mass fraction contours illustrate the reaction dynamics and mixing processes of the non-premixed flame. Methane fuel (CH_4 , Figure 4a) enters through the central nozzle and is quickly consumed upon combustion with the oxygen (O_2 , Figure 4c) from the surrounding annular air co-flow. Because the reaction is driven by rapid mixing within the turbulent shear layer [2], the fuel is completely depleted over a very short axial distance. Simultaneously, the primary combustion products carbon dioxide (CO_2 , Figure 4b) and water vapor (H_2O , Figure 4d) are generated exclusively along the stoichiometric mixing boundary, which is the precise interface where the fuel and oxygen meet in the perfect chemical ratio for complete combustion. The spatial distribution of these products forms an expanding diagonal wedge that reflects the high-temperature hotspots observed in the thermal contours. As the flow moves downstream, the generated products diffuse radially outward and are carried toward the exit. Since the combustor operates with 28% excess air, some unreacted oxygen remains outside the primary reaction zone and is carried into the reactor exhaust.

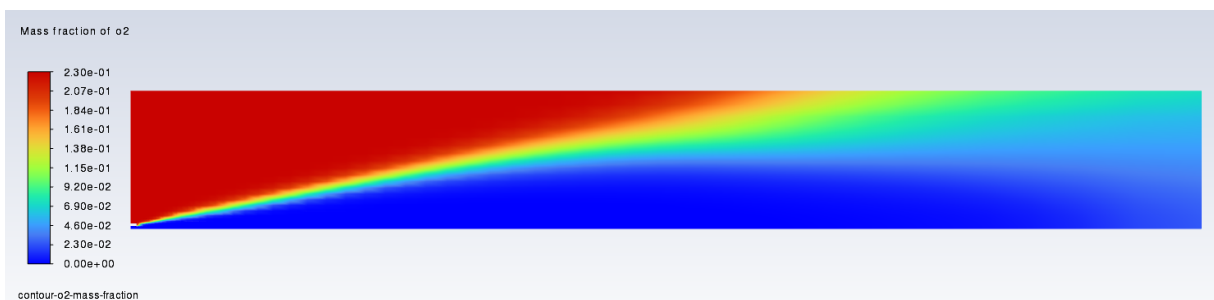
5.1 Contour plots - Task b)



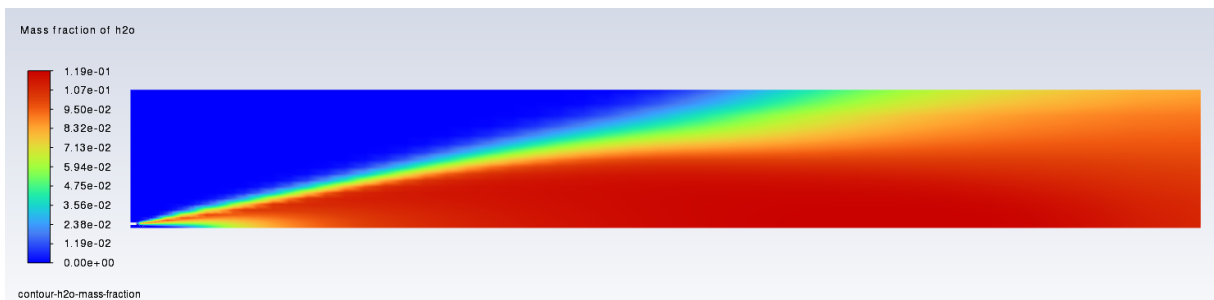
(a) CH₄ is concentrated at the central inlet and is consumed quickly.



(b) CO₂ is formed below the O₂ - CH₄ mixing boundary.



(c) O₂ is consumed above the O₂ - CH₄ mixing boundary, with some excess O₂ remaining at the outlet.



(d) H₂O is formed in the same reaction with CO₂, and shows similar contour plot.

Figure 4: Species mass fractions of (a) CH₄, (b) CO₂, (c) O₂, and (d) H₂O.

5.1.3 Velocity vectors

The velocity vector plot offers a glimpse into the dynamics of momentum transport and the aerodynamic influences of combustion within the reactor. At the inlet boundary on the left, we observe a contrast in the kinematics of the two fluid streams: the central methane jet enters at a velocity of 80 m/s, as highlighted by the red vectors, while the surrounding annular co-flow of air moves at a modest 0.5 m/s. As the methane and oxygen mix and ignite along the shear layer, the significant heat released from this exothermic reaction leads to a dramatic reduction in local gas density [1]. This rapid volumetric expansion accelerates the newly formed high-temperature product species, namely CO_2 and H_2O , propelling the flow radially outward from the flame front. Thus, the evolution of the velocity field emerges not solely from mechanical momentum transfer between the swift jet and the slower co-flow but is predominantly driven by the thermal expansion of the reacting gases [1, 2].

As the flow progresses axially, the high-velocity core experiences a rapid deceleration. This phenomenon is significantly influenced by the intense turbulent shear stresses at the interface of the two streams, which effectively transfer momentum outward from the central jet into the slower-moving bulk fluid [3]. This vital momentum transfer serves as the primary mechanical driver for the turbulent mixing that facilitates the combustion reactions previously discussed. Furthermore, the volumetric expansion from the intense flame heat also propels the flow radially outward. By the time the fluid reaches the exit boundary, the velocity profile has largely equilibrated to a uniform, lower-speed flow. Consistently, the area-weighted average exit velocity of the mixed exhaust gas has been determined to be 3.31 m/s [1, 2].

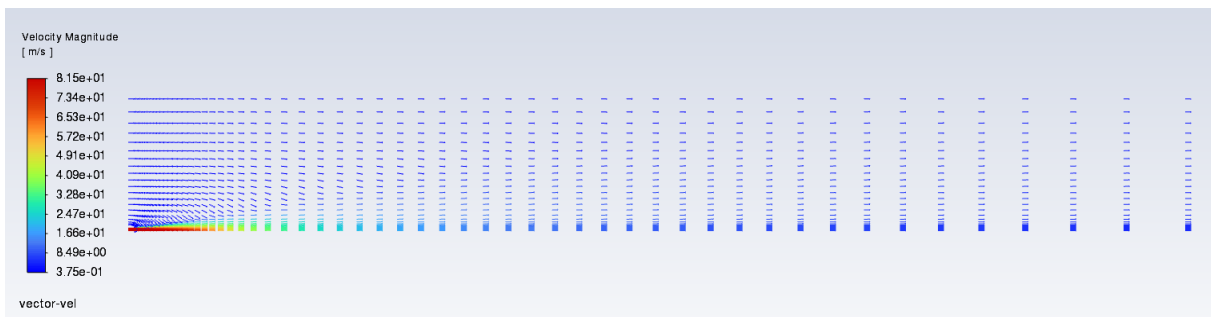


Figure 5: The velocity vector plot, colored by velocity magnitude (m/s), demonstrates fluid velocity profile along the combustor, highlighting the radial momentum transfer into the surrounding low-velocity air co-flow, resulting in uniform exit velocity profile at the outlet.

5.2 Mesh Refinement - Task c)

The computational domain was discretized into quadrilateral cells. Mesh biasing was applied on the inlet and the symmetry axis to ensure smaller control volumes at the regions where large gradients are expected. The minimum orthogonality and maximum skewness of the mesh are 1.0 and 0.0, respectively. Detailed mesh parameters can be found in Appendix 7.

Figure 6 presents the mass-weighted average outlet temperature using coarse, medium and fine mesh resolutions. The quantity ΔT_{fine} shows the absolute and relative temperature error to the fine-mesh result. The coarse mesh, containing 406 cells, deviates from the fine mesh by 7.31 K, corresponding to 0.399 %. When the cell count is quadrupled to 1624 cells with the medium mesh, the deviation is decreased to 2.58 K, or 0.140 %. The fine mesh contains 6 496 cells and resulted in a mass-weighted average outlet temperature of 1833 K.

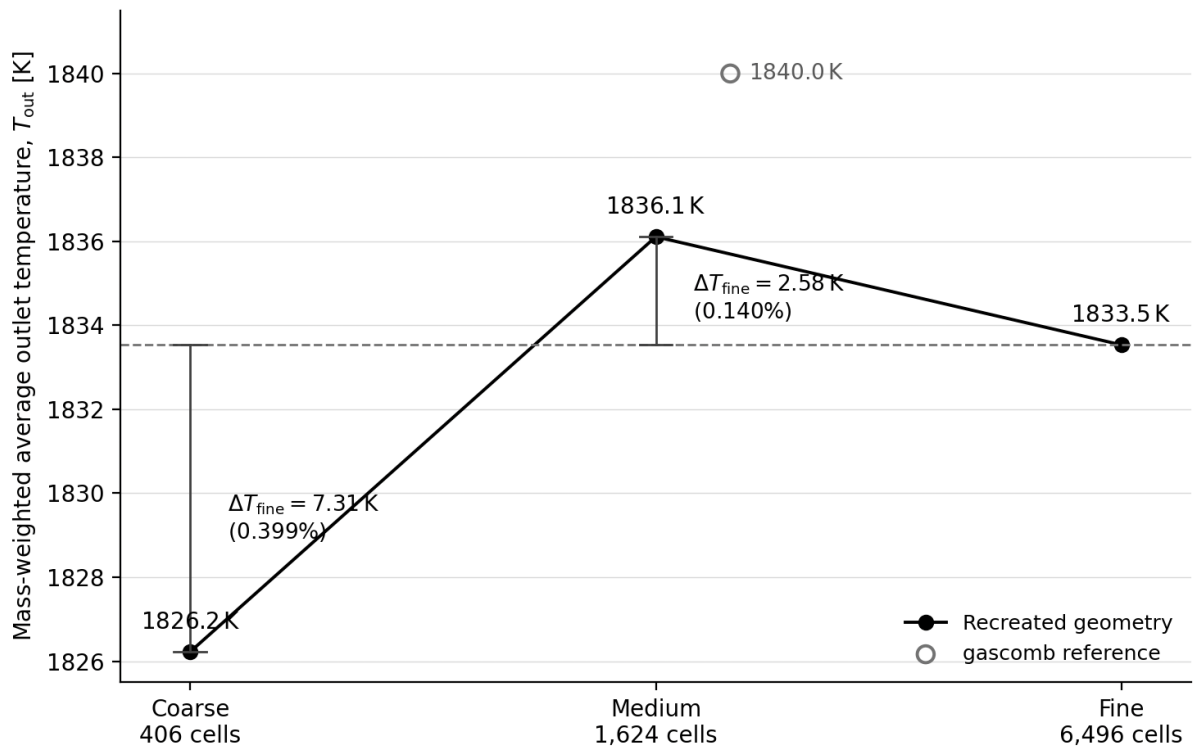


Figure 6: Comparison of mass-weighted average temperature and cell count for recreated geometry and mesh. The error ΔT_{fine} is calculated as the difference between the fine and coarser mesh result. The calculated result for the reference gascomb geometry is marked in grey.

5.3 NO_x - Task d)

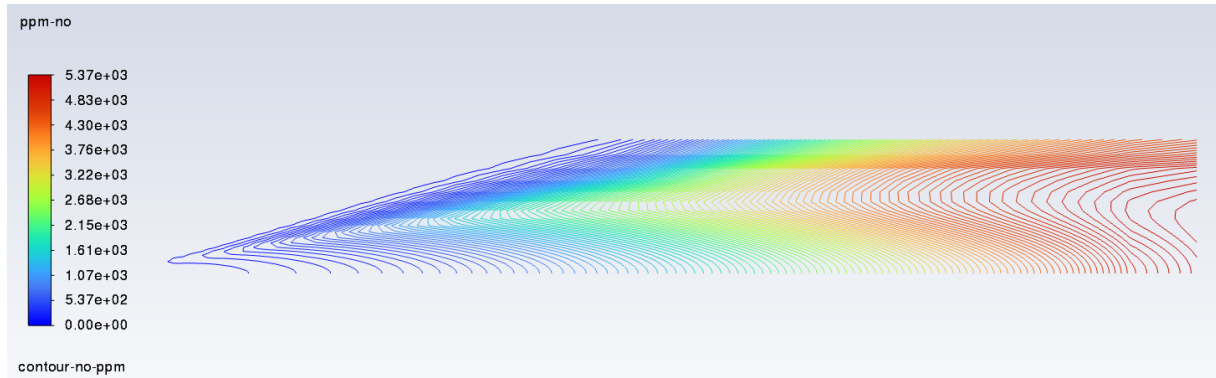
NO production along the combustor is modeled using Fluent thermal and prompt models at an equivalence ratio of 0.76 for prompt model. The thermal model calculates the NO formation via the radical mechanism (See Eqs. 3.4-3.6), while the prompt model includes additional NO-forming mechanisms. The thermal model uses a partial equilibrium approach to predict the oxygen radical concentration [6]. The NO production is simulated after the baseline solution has converged, using the baseline solution as an initial guess. The convergence criteria are set based on temperature variance and NO equation, at 10^{-3} and 10^{-6} , respectively.

The contour plots in Figures 7a-7c show the NO concentration along the axisymmetric combustor in ppm. The concentration in ppm (dry) is calculated as shown in Eq. 5.1 [6], where X_i is the mole fraction of each component. The resulting value is multiplied by 10^6 to convert to ppm.

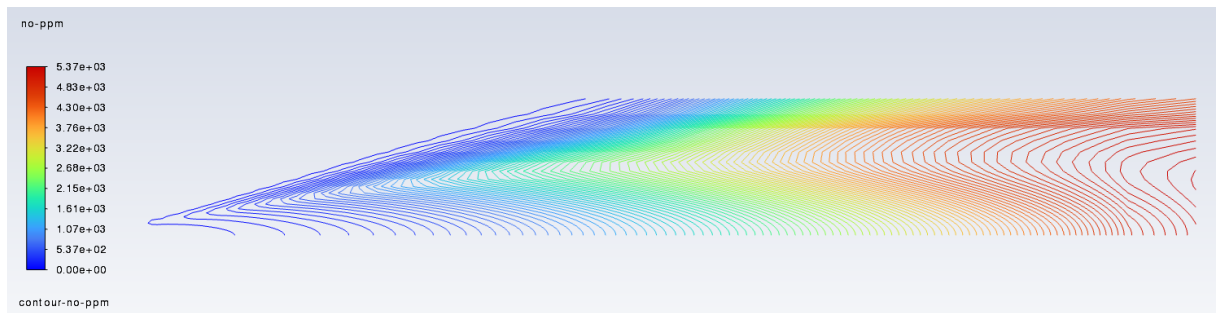
$$\text{NO}_{\text{ppm,dry}} = \frac{X_{\text{NO}}}{1 - X_{\text{H}_2\text{O}}} \times 10^6, \quad (5.1)$$

Most of the NO is produced thermally, with higher concentrations at the end of the combustor allowing more time for NO formation. Prompt NO formation is significantly lower, as shown in Figure 7c. The peak in prompt NO concentration is located earlier in the combustor compared to thermal NO. This is explained by Fenimore [10], who identified the prompt mechanism to describe the faster, transient formation of NO.

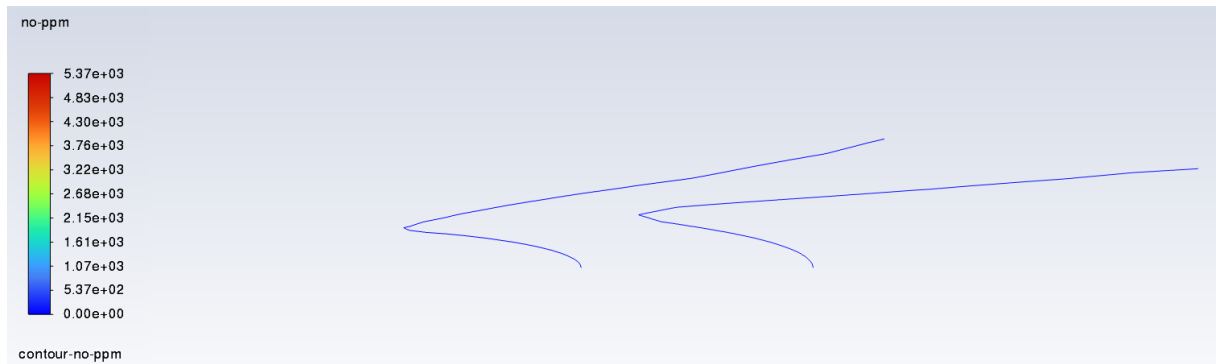
The outlet NO concentration in ppm was calculated using the mass-weighted average surface integral at the outlet of the combustor. The concentration of NO with combined thermal and prompt models is 5067 ppm, with thermal-only resulting in 5028 ppm and prompt-only in 112 ppm. The individual prompt and thermal contributions do not add up, as the reactions are reversible. As a result, some of the product can be destroyed in the backward reaction via another mechanism when both mechanisms are modeled together [6].



(a) Thermal + prompt



(b) Thermal



(c) Prompt

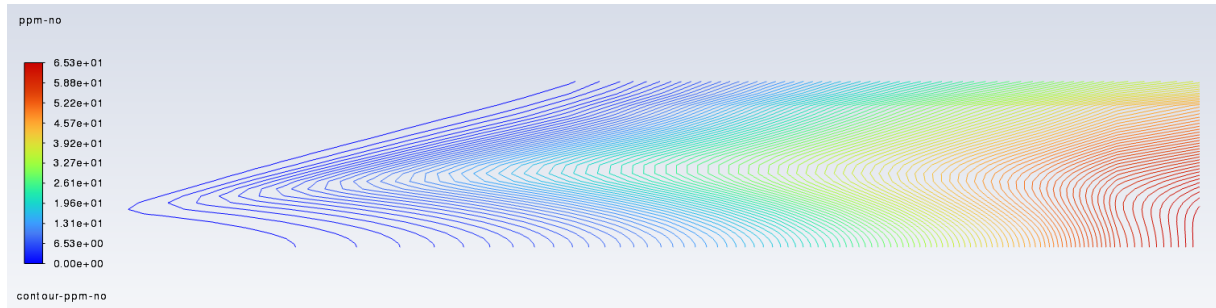
Figure 7: Contour plots showing NO formation through different mechanisms.

5.4 Flue Gas Recirculation (FGR) - Task e) & f)

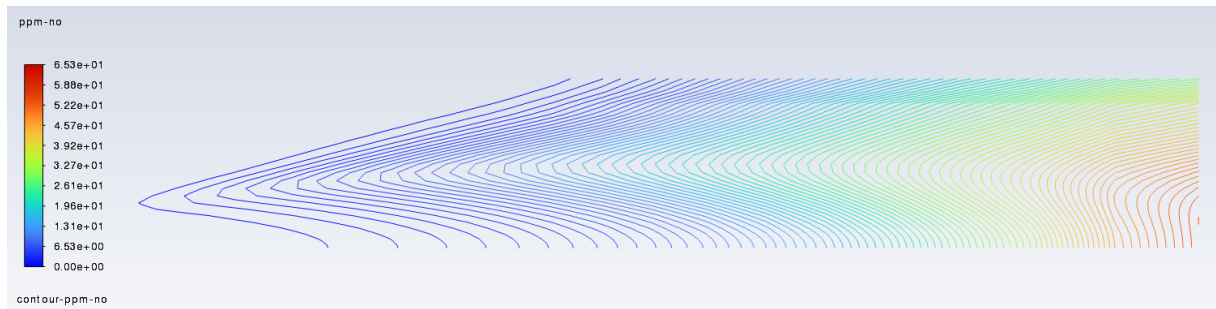
Flue Gas Recirculation (FGR) is an additional stream of fluid that is recirculated back from the combustor outlet to the inlet. FGR results in reduction in combustion reactivity as a consequence of oxygen depletion and increase in CO_2 content [16]. As this influences NO_x formation, FGR can be used to control the NO_x emissions. For this setup, the FGR is modeled with a recycle ratio of 0.25 and inlet velocity of 6 m/s . The recycle ratio was calculated based on the total outlet mass flow, and the FGR inlet radius was calculated to match the recycle ratio and inlet velocity, resulting in FGR radius of 0.038 m. The FGR stream is cooled to 473 K, and the cooling process is excluded from the analysis. The FGR inlet mass fractions were determined based on the composition of the outlet. To match the recycle inlet composition to the outlet, repeated simulations were run and the recycle inlet mass fractions were edited to match the outlet mass fractions. The exact boundary conditions can be found in Table 7 in Appendix.

The contour plots in Figures 8a-8c show the distribution of NO along the combustor with FGR. The similarity of Figures 8a and 8b illustrate how most of the NO is formed via the thermal pathway, with the highest concentrations located at the outlet. The prompt NO shows the highest concentration earlier in the combustor, indicating shorter reaction time [10]. Overall, the NO concentration is significantly lower with all mechanisms throughout the combustor compared to the system without FGR.

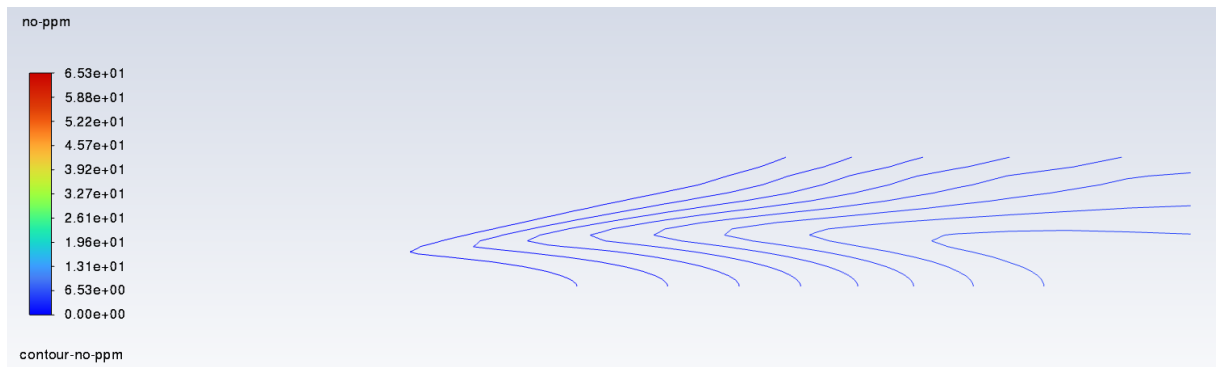
NO production was calculated using mass-weighted average surface integral at the outlet. The combined thermal and prompt models resulted in an outlet concentration of 52 ppm. The thermal and prompt models alone resulted in 48 ppm and 4.5 ppm, respectively. This indicates a nearly 99% reduction in the total thermal and prompt NO emissions when using FGR. The prompt only NO concentration decreased by 96% and thermal only by 99 %.



(a) FGR Thermal + prompt



(b) FGR Thermal



(c) FGR Prompt

Figure 8: Contour plots showing NO formation through different mechanisms with FGR.

5.5 Bayesian Optimization - Additional Task g)

Bayesian Optimization (BO) was used to explore the FGR operating conditions with a limited number of CFD simulations. The goal is to lower the NO production by optimizing the inlet conditions v_{air} , v_{FGR} and T_{FGR} . Details of the temperature constrained BO setup can be found in Appendix 8.

The Gaussian Process regression surrogates (GP) were first fitted to the initial set of 10 data points, as shown in Table 1. The new samples for CFD simulations were chosen one-by-one using the Expected Improvement (EI) algorithm (See Appendix 7). After each sample, the GPs were refitted and a new sample from input space was generated by the EI. This was run for 9 iterations, until there was no substantial improvement in the EI and the new sample locations were in close proximity to the previous samples. This indicates that the EI is no longer exploring new regions.

Most of the EI samples lie in the same region of the input space. One reason for this behaviour can be the setup of the constrained EI, that results in a high EI when the average NO is very low, given the constraint $\mathbb{P}(T \geq 0.95 T_{baseline} \mid \mathcal{D})$ is fulfilled. Since NO is strongly influenced by temperature [17], EI mostly samples around the lower bound of v_{air} and higher bound of v_{FGR} . These conditions lower the oxygen content in the combustor, lowering the temperature and the NO posterior mean, as shown in Fig. 9. As long as the temperature constraint is fulfilled, the EI decreases for samples outside this low NO region. As a result, outside this low NO region EI prefers samples where the uncertainty is high (See Sim. 18 in Tab. 1). Looking at Fig 9, selecting a higher minimum temperature changes the feasible region and can shift the region where EI samples.

Lower oxygen concentration in the combustor also influences the completeness of methane combustion. Tables 1 and 2 column $Y_{CH_4,out}$ show the outlet methane mass fractions. Comparing the CFD data shows that lower $Y_{NO,out}$ is paired with lower methane conversion with a higher $Y_{CH_4,out}$, or T_{out} that is below the temperature requirements (Sim. 10 in Tab. 1). To ensure NO concentration is not lowered by incomplete methane combustion, an additional constraint for outlet methane composition should be implemented analogously to the temperature constraint.

Table 2 shows the next samples sampled after the EI sampling, chosen using the constrained optimization algorithm (See Appendix 7). OS1 is the constrained optimum after fitting the 10 initial samples and 9 EI samples. To test the sensitivity of the optimum to the temperature constraint, the optimum was computed using different constraints $\mathbb{P}(T \geq p \times T_{baseline} \mid \mathcal{D}) \geq 0.95$ with $p \in \{0.95, 0.97, 0.99\}$. The CFD results and error metrics in Table 3 show that increasing T_{min} shifts the constrained optimum towards higher T_{FGR} resulting in higher NO

concentrations, demonstrating the trade-off between high outlet temperature and low outlet NO concentration. The optimum OS1.1 was determined after refitting the GPs with these CFD data sampled at higher temperatures. As shown in Figure 10, the GP prediction relative error at this new optimum is 83.3 % with the NO model and absolute error is 43.36 K with the temperature model. This shows that the optimum is shifted by including nearby points at higher temperatures, and the GPs predict poorly in this region. To verify whether the new optimums are accurate and improving, OS1.2 and OS1.3 were sequentially sampled and fitted after fitting OS1.1. For NO, both predictions are accurate and decreasing from the original optimum OS1. The temperature predictions for OS1.1-OS1.3 lie outside the 1σ uncertainty bounds estimated by the temperature GP. The GP predicted temperature is overestimated for OS1.1 and OS1.3 and the actual CFD result does not fulfill the temperature constraint.

Case OS1.2 contains the optimal inlet conditions $x = [v_{air} = 0.361 \text{ m/s}, v_{FGR} = 9.54 \text{ m/s}, T_{FGR} = 459 \text{ K}]$ resulting in the lowest NO concentration under the temperature constraint with 0.033 ppm outlet NO concentration and 2.73 % relative error to the NO GP posterior mean. OS1.3 CFD outlet temperature is 0.25 K under the temperature constraint. Compared to FGR baseline, the thermal NO was reduced by 99.9 %. This indicates that BO can be used to find an optimum requiring a lower number of expensive CFD simulations, with only 10 initial samples and 13 more samples with targeted active learning algorithms.

Table 1: CFD data used for training and sequential updating of the Gaussian Process surrogate models. Simulation 2 is omitted from BO dataset because the design bounds were subsequently changed after examining the results. The next sample EI 10 is 5.302×10^{-9} , with $v_{air}=0.35 \text{ m/s}$, $v_{FGR}=10.0 \text{ m/s}$ and $T_{FGR} = 482.47 \text{ K}$.

Sim.	Sampling stage	v_{air} [m s ⁻¹]	v_{FGR} [m s ⁻¹]	T_{FGR} [K]	$Y_{NO,out}$ [-]	T_{out} [K]	$Y_{CH_4,out}$ [-]	EI(x) [-]
1	Baseline	0.50000	6.00000	473.15	3.8213×10^{-5}	1618.713	1.9568×10^{-4}	—
3	Initial sample	0.44966	2.53970	428.34	4.3954×10^{-4}	1831.888	—	—
4	Initial sample	0.54215	6.84380	694.67	2.5498×10^{-4}	1643.808	—	—
5	Initial sample	0.41901	9.16760	554.32	3.0964×10^{-6}	1667.976	5.8805×10^{-4}	—
6	Initial sample	0.58920	5.47170	912.39	1.1168×10^{-3}	1636.297	9.9409×10^{-5}	—
7	Initial sample	0.49376	7.04570	941.59	7.0088×10^{-4}	1772.613	1.5472×10^{-4}	—
8	Initial sample	0.51443	3.35010	681.37	1.0088×10^{-3}	1751.232	1.6985×10^{-4}	—
9	Initial sample	0.37493	4.66980	822.04	2.4794×10^{-4}	1959.406	2.9925×10^{-3}	—
10	Initial sample	0.61686	8.97410	457.84	1.2898×10^{-5}	1381.777	3.6427×10^{-5}	—
11	Initial sample	0.43817	8.31530	794.63	1.0928×10^{-4}	1797.677	3.6874×10^{-4}	—
12	EI 1	0.35000	10.00000	373.15	4.7233×10^{-10}	1413.779	4.7022×10^{-3}	—
13	EI 2	0.35000	9.11111	512.45	1.3775×10^{-7}	1599.856	6.2620×10^{-3}	—
14	EI 3	0.35000	9.36569	479.38	4.7900×10^{-8}	1557.106	6.7820×10^{-3}	1.00×10^{-7}
15	EI 4	0.350012	9.46797	465.31	2.9509×10^{-8}	1538.274	6.9627×10^{-3}	2.285×10^{-8}
16	EI 5	0.35000	7.48954	373.15	2.6761×10^{-8}	1530.814	7.0422×10^{-3}	9.579×10^{-9}
17	EI 6	0.350342	9.99088	483.19	2.7154×10^{-8}	1534.718	7.0113×10^{-3}	6.80×10^{-9}
18	EI 7	0.65000	2.00000	373.15	6.6753×10^{-4}	1523.493	1.0907×10^{-4}	4.826×10^{-10}
19	EI 8	0.37489	9.28916	435.52	4.3641×10^{-8}	1567.606	2.2933×10^{-3}	2.623×10^{-9}
20	EI 9	0.35000	10.00000	481.83	2.3806×10^{-8}	1533.810	6.3296×10^{-3}	5.864×10^{-9}

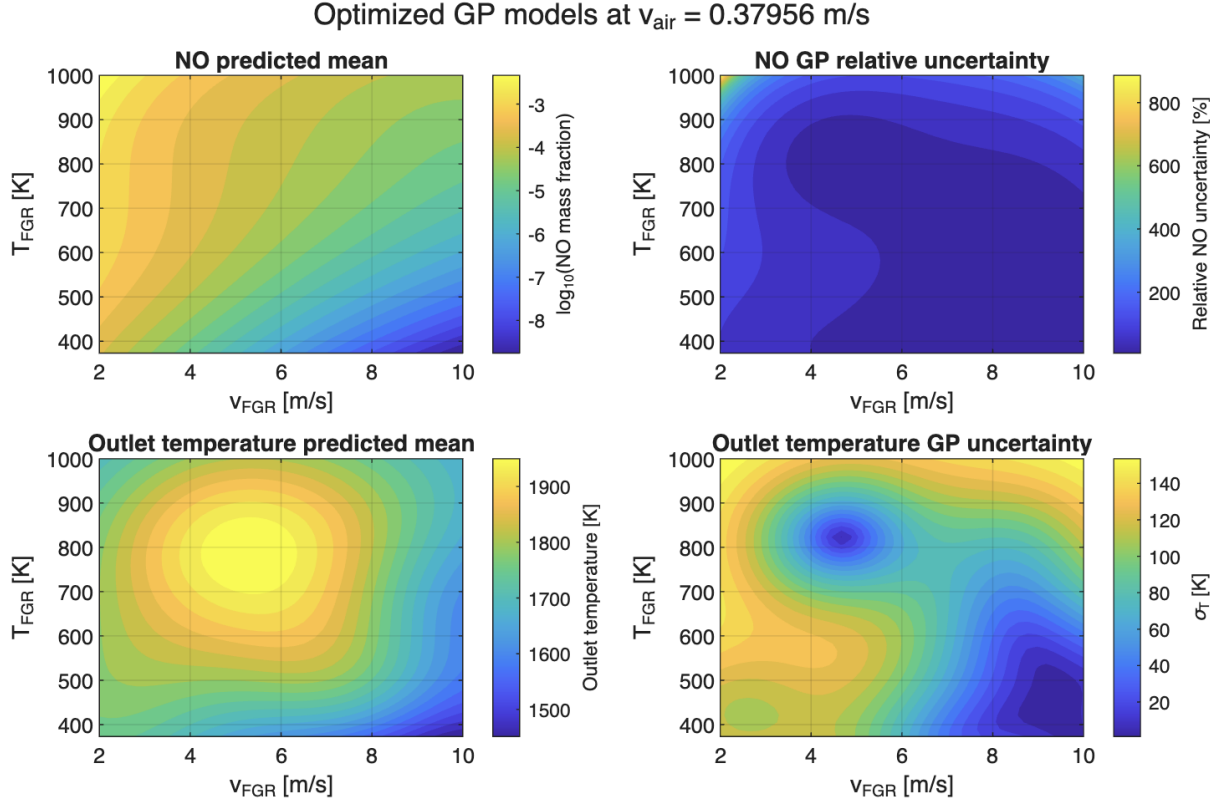


Figure 9: Contour plot of the predicted GP posterior mean and standard deviation for outlet NO (logarithmic) and temperature.

Table 2: CFD verification and refinement of candidate optima for $T_{\min} = 0.95 T_{\text{baseline}}$.

Case	Input conditions			NO				Outlet temperature			$Y_{\text{CH}_4, \text{out}}$ [-]
	v_{air} [m/s]	v_{FGR} [m/s]	T_{FGR} [K]	GP [-]	CFD [-]	CFD [ppm]	Rel. error [%]	GP [K]	CFD [K]	$ e_T $ [K]	
OS1	0.35256	9.5443	472.24	3.335×10^{-8}	3.483×10^{-8}	0.03922	4.25	1546.28	1547.24	0.96	6.59×10^{-3}
OS1.1	0.38942	9.4607	391.09	1.630×10^{-8}	8.893×10^{-9}	0.01003	83.27	1553.67	1510.31	43.36	2.40×10^{-3}
OS1.2	0.36102	9.5394	458.88	3.039×10^{-8}	2.958×10^{-8}	0.03338	2.73	1543.60	1552.76	9.16	4.76×10^{-3}
OS1.3	0.37956	9.3838	414.78	1.728×10^{-8}	1.667×10^{-8}	0.01882	3.69	1539.81	1537.54	2.27	3.01×10^{-3}

Table 3: Sensitivity of the constrained optimum to the minimum outlet temperature requirement.

$T_{\min}/T_{\text{baseline}}$ [-]	$\mathbf{x} = (v_{\text{air}}, v_{\text{FGR}}, T_{\text{FGR}})$	$T_{\min}$ [K]	CFD T_{out} [K]	CFD NO [-]	NO error [%]	$ e_T $ [K]	$Y_{\text{CH}_4, \text{out}}$ [-]
0.95	(0.35256, 9.5443, 472.24)	1537.79	1547.24	3.483×10^{-8}	4.25	0.96	6.59×10^{-3}
0.97	(0.35408, 9.3827, 498.49)	1570.17	1582.65	8.748×10^{-8}	2.59	2.54	6.39×10^{-3}
0.99	(0.35362, 9.1467, 526.09)	1602.54	1616.14	2.081×10^{-7}	13.65	0.35	6.43×10^{-3}

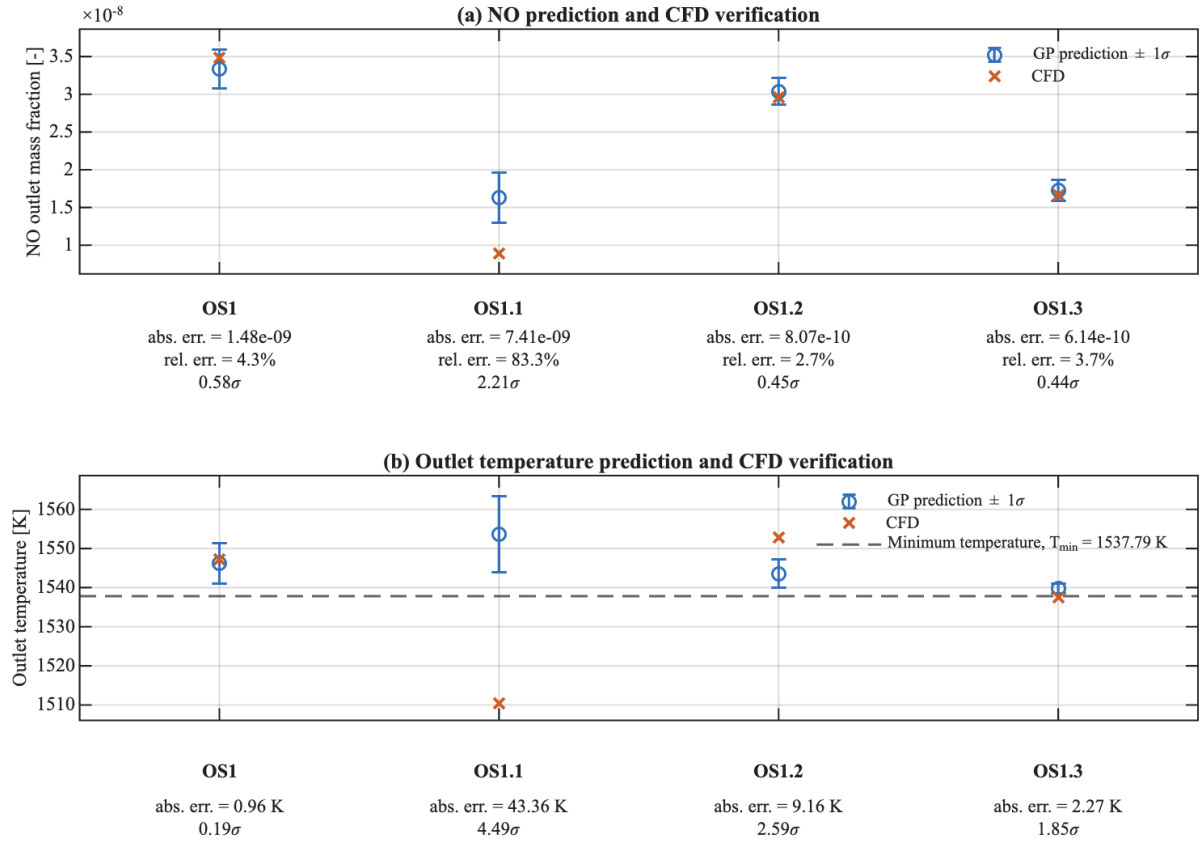


Figure 10: Comparison of Gaussian Process (GP) predictions with the corresponding CFD verification results following the temperature constraint sensitivity analysis. The absolute and relative errors are reported along with the error in terms of GP predicted standard deviation. The errors are computed before fitting each data point to the GP. The dashed line denotes the minimum outlet-temperature requirement $T_{\min} = 0.95 T_{\text{baseline}}$.

6 Discussion

Both the temperature distribution (Fig. 3) and the CO_2 and H_2O (Fig. 4b and 4d) contour plots show a similar color profile, with a near-diagonal abrupt change in temperature and species concentrations. As the methane mixes with oxygen, exothermic combustion produces heat, water and CO_2 , while fast fuel-breakdown reactions quickly consume the methane. Since turbulent flow and expansion of combusted gases enhance mixing of the reactant species, this region of high temperature and high combustion product concentrations is distributed annularly towards the combustor walls [18][19]. The effect of mesh resolution on the computed temperature is minimal, with an improvement from 0.399 to 0.14 % in relative error in respect to the fine mesh mass-weighted average outlet temperature. This supports using the recreated mesh for NO_x modeling.

Comparison of the contour plots for NO (Fig. 7a) and other combustion products (Fig. 4b for CO_2 and Fig. 4c for O_2) shows that NO requires longer reaction time compared to methane combustion. The kinetics of NO formation are slow because of the triple bond in N_2 that is broken [17]. In contrast, combustion of CH_4 and O_2 through the formaldehyde intermediate is much faster [1]. As the calculations were performed for only CO_2 although the combustion happens through the CO intermediate [20], not including CO can lead to overpredicted CO_2 concentrations if all the CO does not form CO_2 .

Implementing FGR reduced the mass-weighted average outlet NO concentration from 5067 to 52 ppm. The thermal and prompt NO outlet concentrations reduced from 5028 to 48 ppm and 112 to 4.5 ppm, respectively. The recycle stream consists of nitrogen and combustion products from the outlet, reducing the amount of oxygen available for combustion [16]. This influences both prompt NO by reducing the available oxygen for NO formation and thermal NO by lowering temperature due to reduced available oxygen combustion. Since thermal NO formation is significantly influenced by temperature [17], thermal NO concentration is significantly lower with FGR. The equivalence ratio is set to 0.76 for prompt NO, but this value is not corrected for the actual equivalence ratio with the FGR. Prompt NO concentration decreases relatively less, ($\approx 96\%$) compared to thermal NO ($\approx 99\%$).

Overall, the results indicate that the total thermal and prompt NO pollution can be reduced up to 99 % with FGR, although the influence on combustion efficiency due to lower temperature was not studied. Further optimization of the inlet conditions reduced thermal NO production even further, by 99 % to 0.033 ppm in respect to the FGR baseline. Despite the reduction in temperature and methane conversion, BO presents as a promising method to optimize the system inputs for the desired conditions. While absolute values of the results should be used as an estimate, the general trends are consistent with other research, supporting the use of BO and CFD simulations to investigate low-emission combustion systems.

7 Conclusion

This study simulated the steady-state turbulent combustion of methane in a 2D axisymmetric cylindrical furnace using ANSYS Fluent. By utilizing a pressure-based solver, the $k - \omega$ SST turbulence model, and the Eddy-Dissipation Model for species transport, the interactions among fluid dynamics, heat transfer, and chemical kinetics were captured through contour plots of velocity vectors, temperature and species concentrations. The results indicate that rapid turbulent mixing at the shear layer between the high-velocity methane jet and the slower air co-flow creates a well-defined flame front. This primary reaction zone is characterized by peak temperatures approaching 2300 K, the immediate generation of combustion products (CO_2 and H_2O), and significant radial flow expansion driven by exothermic heat release. Additionally, a mesh refinement study showed numerical stability; the medium mesh, consisting of 1,624 cells, showed a negligible 0.140% temperature deviation relative to the fine mesh. This ensures a computationally efficient yet accurate spatial discretization for subsequent emission modeling.

In terms of emissions, the baseline nitrogen oxides (NO) formation was predominantly influenced by the temperature-dependent thermal (Zeldovich) mechanism, resulting in an exit concentration of approximately 5067 ppm. The implementation of Flue Gas Recirculation (FGR) proved to be a highly effective emission control strategy, reducing total NO emissions by nearly 99%, to about 52 ppm. This significant reduction was achieved because the recirculated exhaust gases dilute the available oxygen and lower peak combustion temperatures, effectively suppressing both the thermal and prompt NO formation pathways. The NO production was further reduced by optimizing the FGR inlet conditions $x = [v_{air}, v_{FGR}, T_{FGR}]$ using Bayesian Optimization. Using 10 initial data points followed by 9 points generated with Expected Improvement (EI) sampling and 4 data points sampled at the optimum for testing the sensitivity of the optimum, the outlet thermal NO concentration was reduced by 99 % to 0.033 ppm. Overall, this computational fluid dynamics (CFD) analysis underscores FGR as a crucial design strategy for lowering the NO_x emissions in high-temperature combustion systems. With 23 additional CFD simulations, BO can be used to further optimize the system.

Project Contributions

The baseline CFD project tasks were completed collaboratively by the group. The Bayesian Optimization extension, including the Gaussian Process surrogate modeling, constrained Expected Improvement sampling, optimization of the FGR operating conditions, and CFD verification of the predicted optima, was developed and implemented by **Kiia Kaarevirta**.

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Appendix

A.1 Mesh Parameters and Quality

Table 4: Mesh parameters and quality metrics for the gascomb baseline mesh, the three meshes used in the mesh-independence study, and the FGR geometry. All horizontal edges used a bias factor of 35 with soft behaviour, whereas the vertical edges used hard behaviour. Face meshing with quadrilateral elements was applied. All remaining settings used the default CFD meshing preferences. The gascomb geometry contains an empty space with length of 3 cells between the methane and air inlets. After this, the vertical cell count totals at 28 cells.

Parameter	Gascomb	Coarse	Medium	Fine	FGR
Total cell count	1615	406	1624	6496	1624
Air inlet cell count	20	10	20	40	15
Air inlet bias factor	<i>n.a.</i>	8.0	8.0	8.0	2.0
Methane inlet cell count	5	4	8	16	8
Recycle inlet cell count	-	-	-	-	5
Recycle inlet bias factor	-	-	-	-	1.3
Horizontal axis cell count	58	29	58	116	58
Minimum orthogonal quality	1.0	1.0	1.0	1.0	1.0
Maximum skewness	$\approx 10^{-10}$	$\approx 10^{-10}$	$\approx 10^{-10}$	$\approx 10^{-10}$	$\approx 10^{-10}$

A.2 Fluent Input Summary

Table 5: Summary of the principal ANSYS Fluent models and numerical settings.

Setting	Selection
Solver	Pressure-based, steady-state, double precision
Computational domain	Two-dimensional axisymmetric
Turbulence model	SST $k-\omega$
Energy equation	Enabled
Species model	Species Transport with volumetric reactions
Reacting mixture	CH ₄ , O ₂ , CO ₂ , H ₂ O, N ₂
Combustion model	Eddy-dissipation model
Density model	Incompressible ideal gas
Pressure-velocity coupling	Coupled
Pressure discretisation	Second order
Momentum, turbulence, species and energy	Second-order upwind
Initialisation	Hybrid initialisation
Radiation	Not considered

Table 6: Principal boundary conditions used for the baseline combustion case.

Boundary	Type	Flow condition	Additional settings
Methane inlet	Velocity inlet	80 m s ⁻¹	Pure CH ₄ ; turbulence intensity 10%; hydraulic diameter 0.010 m
Air inlet	Velocity inlet	0.5 m s ⁻¹	$Y_{O_2} = 0.23$, balance N ₂ ; turbulence intensity 10%
Outlet	Pressure outlet	0 Pa gauge	Backflow composition corresponding to air
Outer wall	Stationary wall	No slip	Constant wall temperature of 300 K
Axis	Axis	–	Axisymmetric centreline condition

Table 7: Modified and additional inlet boundary conditions for the FGR configuration.

Parameter	Value	Description
Boundary type	Velocity inlet	Additional inlet representing cooled re-circulated flue gas
FGR inlet velocity	6 m s^{-1}	Uniform velocity normal to the inlet
FGR inlet temperature	473.15 K	Assigned cooling temperature of the re-circulated stream
Turbulence intensity	10%	Intensity and hydraulic-diameter specification
Hydraulic diameter	0.076 m	Hydraulic diameter assigned to the FGR inlet
CH ₄ mass fraction	0.0001766	Obtained from the baseline outlet composition
O ₂ mass fraction	0.04796	Obtained from the baseline outlet composition
CO ₂ mass fraction	0.1183	Obtained from the baseline outlet composition
H ₂ O mass fraction	0.09688	Obtained from the baseline outlet composition
N ₂ mass fraction	0.7366	Balance species calculated so that $\sum_i Y_i = 1$
Air-inlet hydraulic diameter	0.364 m	Reduced from 0.44 m due to the additional FGR inlet region

A.3. Bayesian Optimization (BO)

Bayesian Optimization setup

Bayesian optimization (BO) uses a probabilistic surrogate model to represent the objective function over input space $\mathcal{X}$ [21]. In this work, Gaussian Process (GP) regression surrogate model provides the posterior prediction at new test inputs $\mathbf{x}_*$. For fixed hyperparameters, Gaussian prior and Gaussian statistical model the GP posterior has a closed form solution, determined by the assumed zero-mean function $\mu(\mathbf{x}) = 0$ and covariance kernel $k(\mathbf{x}_1, \mathbf{x}_2)$ (See Eq. 7.1) [21].

$$k(\mathbf{x}, \mathbf{x}') = \gamma^2 \exp\left(-\frac{1}{2l^2} \|\mathbf{x} - \mathbf{x}'\|_2^2\right), \quad (7.1)$$

The diagonal of the posterior covariance matrix quantifies the posterior uncertainty in the GP prediction at the test inputs, conditioned on the training data and hyperparameters [21]. The uncertainty from hyperparameter (γ , l and σ_n) selection is not included in the quantified uncertainty σ in this work.

The statistical model is shown in Equation 7.2, where y_i is the training data and $f(\mathbf{x}_i)$ is the unknown true system response to input data $\mathbf{x}_i$. The output y_i includes independent and identically distributed (iid) Gaussian noise ϵ_i , modeled with zero mean and σ_n^2 noise variance. The noise standard deviation σ_n is an optimized hyperparameter.

$$y_i = f(\mathbf{x}_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_n^2), \quad (7.2)$$

Using the statistical model in Eq. 7.2, a Gaussian Process Regression (GP) surrogate model conditioned on the training data can be defined as shown in Eq. 7.3,

$$\mathbf{f}_* \mid \mathbf{X}, \mathbf{y}, \mathbf{X}_*, \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*), \quad (7.3)$$

where $\boldsymbol{\mu}_*$ and $\boldsymbol{\Sigma}_*$ are the posterior mean and covariance, as shown in Eq 7.4 and 7.5, respectively.

$$\boldsymbol{\mu}_* = \mathbf{K}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (7.4)$$

$$\Sigma_* = \mathbf{K}_{**} - \mathbf{K}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{K}_*. \quad (7.5)$$

A GP assumes any finite set of random variables generated with the GP follow a Gaussian multivariate distribution [21]. Therefore, the posterior mean $\boldsymbol{\mu}_*$ and covariance Σ_* are derived from the multivariate Gaussian joint distribution, shown in Eq. 7.6.

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{K}_y = \mathbf{K} + \sigma_n^2 \mathbf{I} & \mathbf{K}_* \\ \mathbf{K}_*^T & \mathbf{K}_{**} \end{bmatrix} \right), \quad (7.6)$$

$\{\sigma_n, l, \gamma\} = \boldsymbol{\theta}$ are hyperparameters that are optimized by maximizing the log marginal likelihood, shown in Eqs. 7.7 and 7.8. Since the optimized hyperparameters $\boldsymbol{\theta}^*$ are fixed in the posterior prediction (See Eq. 7.4 and 7.5), the uncertainty from hyperparameter optimization is not included in the quantified posterior uncertainty.

$$\log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) = -\frac{1}{2} \mathbf{y}^T \mathbf{K}_y^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K}_y| - \frac{N}{2} \log(2\pi), \quad (7.7)$$

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} \log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}). \quad (7.8)$$

Optimization

The goal of the optimization is to find operating conditions $\mathbf{x} = [\mathbf{v}_{air}, \mathbf{v}_{FGR}, \mathbf{T}_{FGR}] \in \mathcal{X}$ resulting in lower NO emissions. Since NO production increases at high temperatures and high oxygen concentrations, and FGR influences both oxygen concentration through dilution and temperature [16], $\mathbf{v}_{FGR}$ and $\mathbf{T}_{FGR}$ are included as inputs. However, lower temperature and lower oxygen concentration can decrease methane combustion completeness. To ensure sufficient temperature with lower NO formation, a probabilistic temperature constraint is imposed in the optimization objective. Temperature is also influenced by oxygen concentration through oxygen availability for combustion, hence $\mathbf{v}_{air}$ is also included as optimization input to control oxygen content in the combustor alongside $\mathbf{v}_{FGR}$.

The constrained optimization problem is shown in Equation 7.9,

$$\mathbf{x}^{\text{opt}} = \arg \min_{\mathbf{x} \in \mathcal{X}} \mu_{\text{NO}}(\mathbf{x}) \quad \text{s.t.} \quad \mathbb{P}(T_{\text{out}}(\mathbf{x}) \geq 0.95 T_{\text{baseline}} \mid \mathcal{D}) \geq 0.95. \quad (7.9)$$

where μ_{NO} is GP posterior mean of the mass-weighted-average (MWA) outlet NO mass fraction, T_{out} is the MWA outlet temperature and T_{baseline} is 1618 K that is the MWA at the baseline conditions at 0.5 m/s v_{air} , 6 m/s v_{FGR} and 473 K T_{FGR} . The predicted mean of the GP surrogate trained on T_{out} is used in the probabilistic temperature constraint $\mathbb{P}(T_{\text{out}}(\mathbf{x}) \geq 0.95 T_{\text{baseline}})$, that ensures the outlet temperature is 95 % of the baseline temperature. The outer inequality ensures the confidence of this prediction is above 95 %. The second GP surrogate trained on NO concentration is used to predict μ_{NO} .

Data processing

The training outputs NO mass fraction and temperature are standardized, allowing for assuming zero-mean for the GP. The outputs are standardized using Eq. 7.10, where μ_y and σ_y are the mean and standard deviation of the training outputs.

$$z = \frac{y_i - \mu_y}{\sigma_y}, \quad (7.10)$$

To constrain the GP to only predict non-negative NO mass fractions, NO mass fraction is logarithmically transformed before standardization and fitting the GP. Therefore, the predicted posterior NO mean μ_* and standard deviation σ_* are also standardized and in log space. They are unstandardized using Eq. 7.11,

$$\mu_{\log} = \bar{y}_{\log} + s_{\log} \mu_*, \quad \sigma_{\log} = s_{\log} \sigma_*. \quad (7.11)$$

where $\bar{y}_{\log}$ and $s_{\log}$ are the mean of the log transformed NO mass fractions and the standard deviation of the log transformed NO training data, respectively. Finally, the mean and standard deviation are transformed from log space to linear space using the log-normal distribution transformations for Gaussians as shown in Equations 7.12 and 7.13.

$$\mu_{\text{NO}} = \exp \left(\mu_{\log} + \frac{1}{2} \sigma_{\log}^2 \right), \quad (7.12)$$

$$\sigma_{\text{NO}} = \sqrt{(\exp(\sigma_{\log}^2) - 1) \exp(2\mu_{\log} + \sigma_{\log}^2)}. \quad (7.13)$$

Inputs $\mathbf{x} = [v_{\text{air}}, v_{\text{FGR}}, T_{\text{FGR}}]$ are normalized, using Eq. 7.14, where $x_j^{\min}$ and $x_j^{\max}$ correspond to the bounds of the optimized input parameters $\mathbf{x}$ (See Table 8). As a result, all input parameters are on a comparable scale, ensuring more consistent optimization.

$$x_{i,j}^{\text{norm}} = \frac{x_{i,j} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}, \quad (7.14)$$

To ensure correct FGR inlet composition with the changing input conditions, the simulation is run a few times while matching the combustor outlet and FGR inlet compositions. The mass weighted average outlet temperature is calculated first without the NO model. After recording the temperature, the simulation is run again with the thermal NO model. Prompt NO was not included, since the oxygen content in prompt NO is influenced by the equivalence ratio. When changing the inlet air and FGR flows, the equivalence ratio should be corrected accordingly. For simplification, only thermal NO is considered.

Initially 9 samples are generated using Sobol sampling. The GPs are fitted to the FGR baseline solution and these 9 data points (initial fit with 10 data points). The bounds were chosen to symmetrically lie around the baseline solution for the fluid velocities. For temperature, physically plausible bounds were selected between 100 and 727 $^{\circ}\text{C}$. The air velocity and FGR velocity are sampled from 0.35 to 0.65 m/s and 2.0 to 10.0 m/s , respectively.

Expected Improvement sampling

Expected Improvement (EI) is used to select the new input conditions where to run new CFD simulations to explore the conditions resulting in the lower NO production with temperature constraint. This approach is commonly used with BO, as simply finding the minimum does not acknowledge the uncertainty about the predictions and can often lead to a local minimum. In contrast, EI balances exploration and exploitation by favoring samples where either the prediction is good or where the uncertainty is high [22]. The expression is shown in Eq. 7.15, where $f_{\min}$ is the current lowest NO concentration that fulfills the temperature constraint $T \geq 0.95 T_{\text{baseline}}$ and f^* is the new predictive NO concentration at new test input conditions.

$$\text{EI}_{\text{NO}}(\mathbf{x}) = \mathbb{E}[\max(f_{\min} - f^*(\mathbf{x}), 0)], \quad (7.15)$$

Since the GP is trained for logarithmic NO mass fractions to avoid negative predictions, the corresponding logarithmic form of EI is used. The derivation of the logarithmic EI follows from Watanabe [23] (Preprint). The full expression using the logarithmic NO GP mean $\mu_{\log \text{NO}}(\mathbf{x})$ and variance $\sigma_{\log \text{NO}}^2(\mathbf{x})$ and the normal CDF evaluated around $z = \frac{\ln(f_{\min}) - \mu_{\log \text{NO}}(\mathbf{x})}{\sigma_{\log \text{NO}}(\mathbf{x})}$.

$$\text{EI}_{\log \text{NO}}(\mathbf{x}) = f_{\min} \Phi(z) - \exp\left(\mu_{\log \text{NO}}(\mathbf{x}) + \frac{1}{2}\sigma_{\log \text{NO}}^2(\mathbf{x})\right) \Phi(z - \sigma_{\log \text{NO}}(\mathbf{x})), \quad (7.16)$$

Assuming statistically independent response variables (i.e. NO mass fraction and temperature), the acquisition function with temperature constraint simplifies to the expression in Eq. 7.17 [24].

$$\alpha(\mathbf{x}) = \text{EI}_{\log \text{NO}}(\mathbf{x}) \mathbb{P}(T_{\text{out}}(\mathbf{x}) \geq 0.95 T_{\text{baseline}} \mid \mathcal{D}). \quad (7.17)$$

512 candidate points for EI are sampled with Sobol sampling inside the training bounds $v_{\text{air}} = [0.35, 0.65]$ m/s, $v_{\text{FGR}} = [2.0, 10.0]$ m/s and $T_{\text{FGR}} = [373, 1000]$ K, with 12 restarts. This ensures global coverage of the optimization surface while having continuous input parameter space because of using optimization with restarts. The local optimization was performed with MATLAB fmincon that is used for constrained nonlinear multivariable minimization problems. Similar approach was used by Lima et al. with a similar system and optimization objective [25] (Preprint). The same sampling approach is also used to find the constrained optimum.

Table 8: Main parameters used for the Bayesian optimization.

Parameter	Value / range	Unit
<i>Design space</i>		
Air inlet velocity, v_{air}	$0.35 \leq v_{\text{air}} \leq 0.65$	m/s
FGR inlet velocity, v_{FGR}	$2.0 \leq v_{\text{FGR}} \leq 10.0$	m/s
FGR inlet temperature, T_{FGR}	$373.15 \leq T_{\text{FGR}} \leq 1000$	K
<i>Temperature constraint</i>		
Baseline outlet temperature, T_{baseline}	1618.7	K
Minimum outlet temperature, $T_{\min}$	$0.95 T_{\text{baseline}} \approx 1538$	K
Minimum feasibility probability, $p_{\min}$	0.95	—